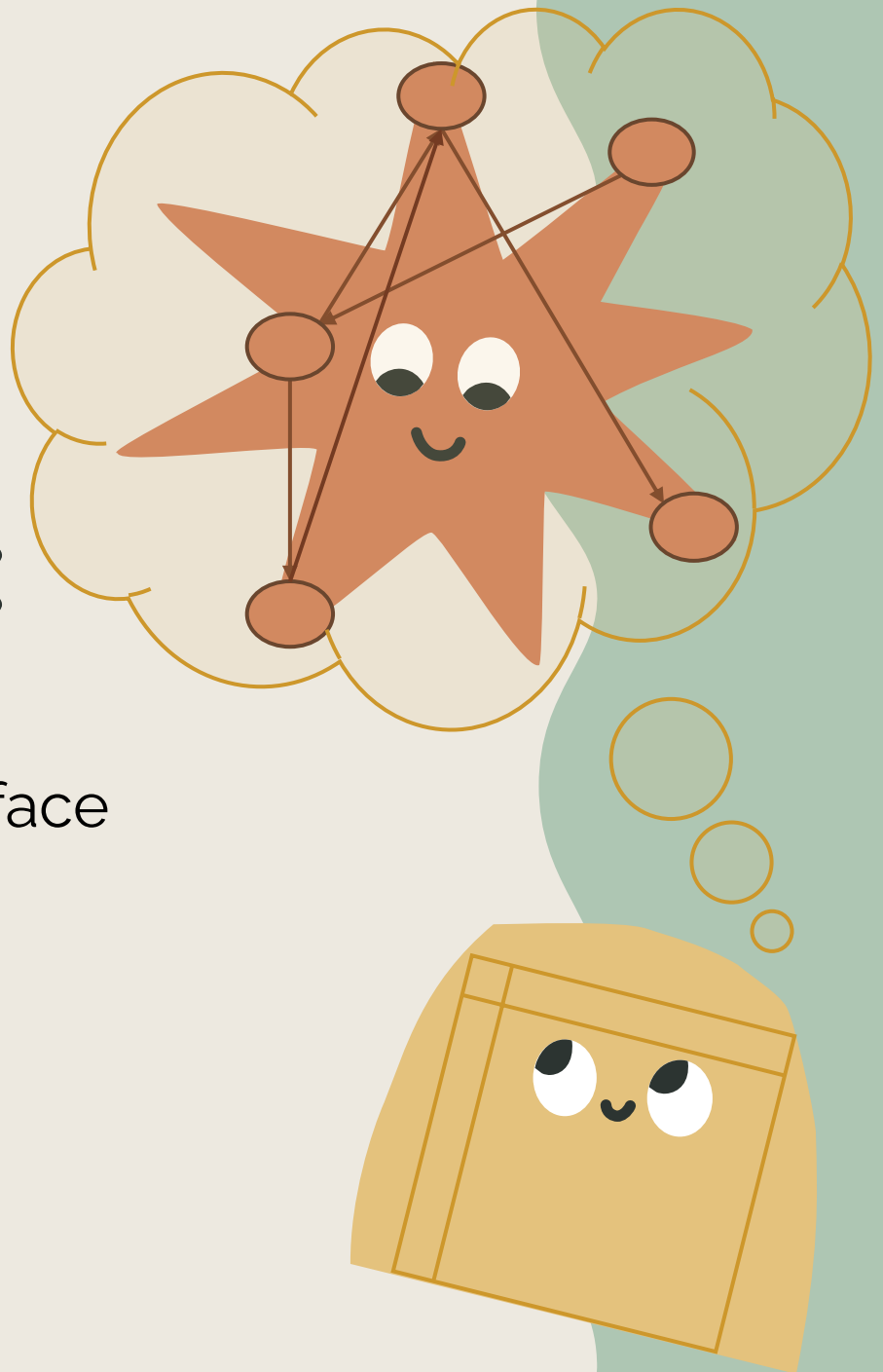
From SHACL constraints to Form-Based RDF Authoring: an RML use case

Turning a specification into a user interface



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Masters student in Computer Science and Engineering at ULiège,
Faculty of Applied Science

LIÈGE université
Sciences Appliquées

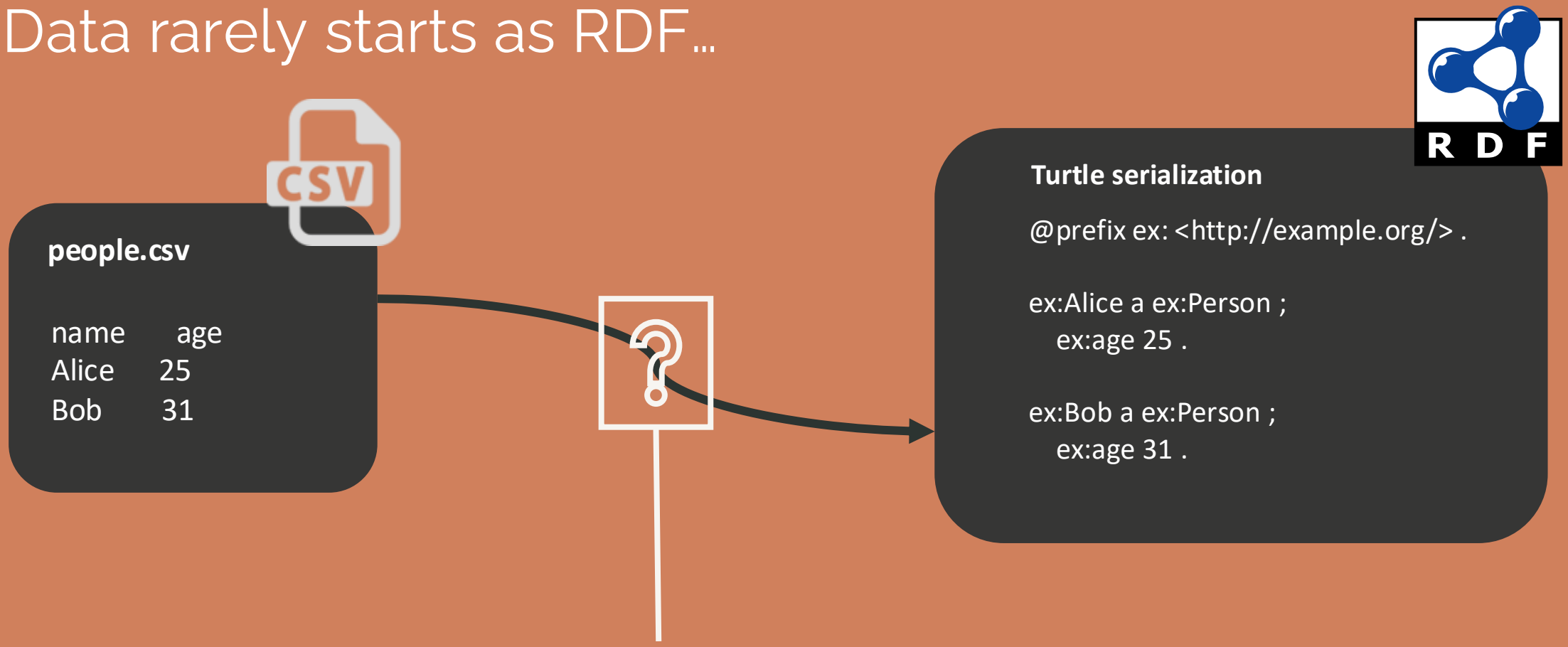
The Problem

Data rarely starts as RDF…



The Problem

Data rarely starts as RDF…



…**How** do we describe the transformation?

RDF, a data model for KG

W3C Recommendation

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Abstract

The Resource Description Framework (RDF) is a framework for representing information in the Web. This document defines an abstract syntax (a data model) which serves to link all RDF-based languages and specifications. The abstract syntax has two key data structures: RDF graphs are sets of subject-predicate-object triples, where the elements may be IRIs, blank nodes, or datatyped literals. They are used to express descriptions of resources. RDF datasets are used to organize collections of RDF graphs, and comprise a default graph and zero or more named graphs. RDF 1.1 Concepts and Abstract Syntax also introduces key concepts and terminology, and discusses datatyping and the handling of fragment identifiers in IRIs within RDF graphs.

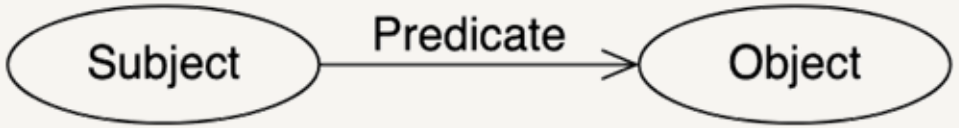


Fig. 1 An RDF graph with two nodes (Subject and Object) and a triple connecting them (Predicate)

Source: RDF 1.1 Concepts and Abstract Syntax. W3C Recommendation. World Wide Web Consortium (W3C), Feb. 2014. url:https://www.w3.org/TR/rdf11-concepts/

The bridge

RDF Mapping Language (RML)

Abstract

This document describes RML, a generic mapping language, based on and extending R2RML. The RDF Mapping language (RML) is a mapping language defined to express customized mapping rules from heterogeneous data structures and serializations to the RDF RDF-CONCEPTS data model. RML is defined as a superset of the W3C-standardized mapping language R2RML, aiming to extend its applicability and broaden its scope, adding support for data in other structured formats. R2RML is the W3C standard to express customized mappings from relational databases to RDF. RML follows exactly the same syntax as R2RML; therefore, RML mappings are themselves RDF graphs. The present document describes the RML-Core language and its concepts through definitions and examples.

Source: RDF Mapping Language (RML) url:https://rml.io/specs/rml/

The Problem

people.csv

name	age
Alice	25
Bob	31

Turtle serialization

```
@prefix rml: <http://w3id.org/rml/> .
@prefix ex:  <http://example.org/> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .

ex:PersonMapping a rml:TriplesMap ;
   rml:logicalSource [
      rml:source [ a rml:RelativePathSource ;
                   rml:root rml:MappingDirectory ;
                   rml:path "people.csv" ] ;
      rml:referenceFormulation rml:CSV
   ] ;
   rml:subjectMap [
      rml:template "http://example.org/{name}" ;
      rml:class ex:Person
   ] ;
   rml:predicateObjectMap [
      rml:predicate ex:age ;
      rml:objectMap [
         rml:reference "age" ;
         rml:datatype xsd:integer
      ]
   ] .
```

Turtle serialization

```
@prefix ex:
<http://example.org/> .

ex:Alice a ex:Person ;
   ex:age 25 .

ex:Bob a ex:Person ;
   ex:age 31 .
```

carml / carml
A pretty sweet RML engine, for RDF.
Java 114 22

morph-kgc / morph-kgc
...DF knowledge graph generation with RML
300 61 Updated 3 weeks ago

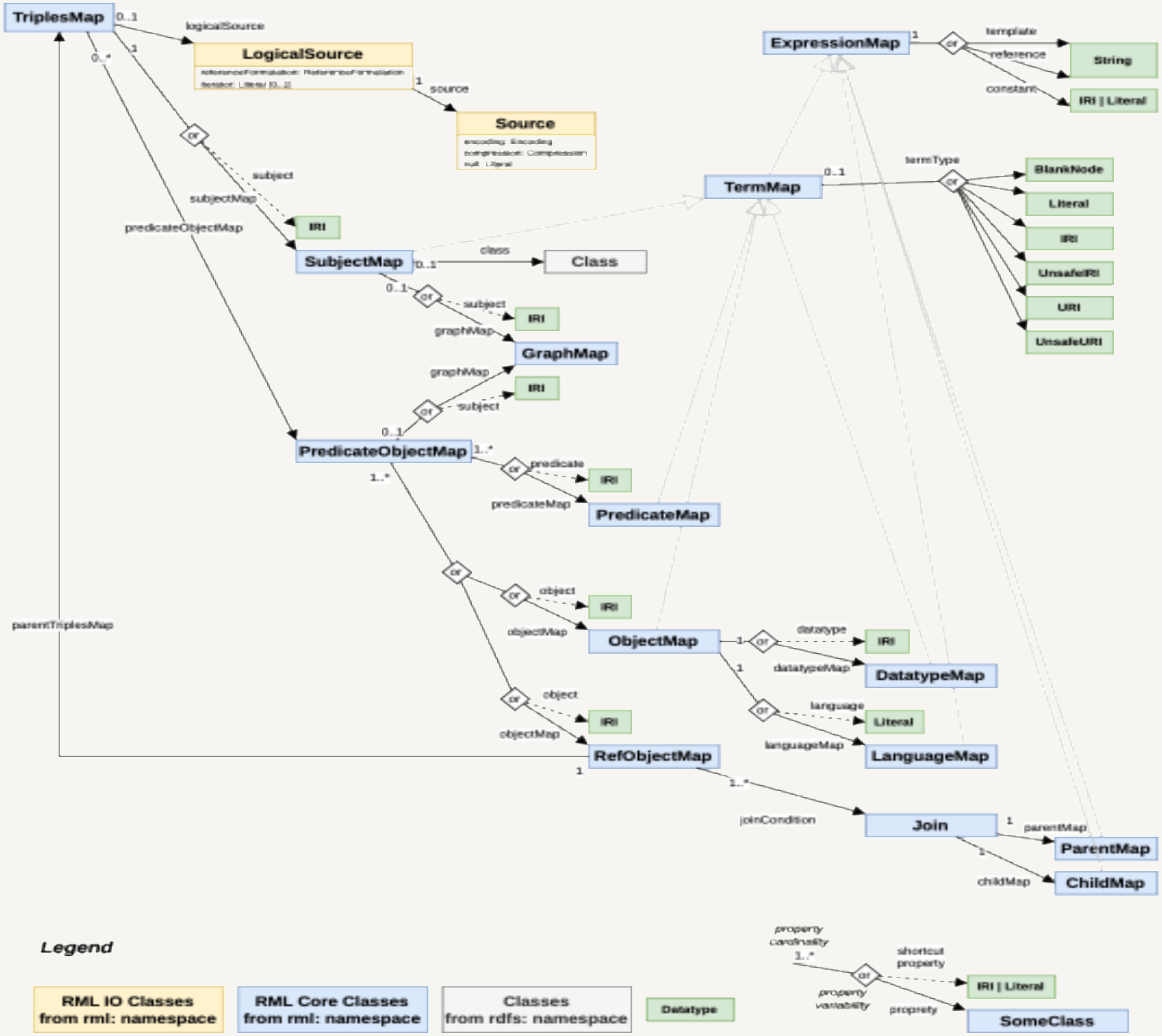
RMLio / rmlmapper-java
The RMLMapper executes RML rules to generate high quality Linked Data from multiple originally (semi-)structured data sources
Java

anuzzolese / pyrml
pyRML is a Python based er...
The RDF Mapping Language...
defined to express customi...
heterogeneous data structu...
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A Basic and Unassuming RML Processor
Java 7 2 Updated 10 hours ago

...**Who** will describe the transformation?

RML ontology



SHACL as the bridge

SHApe Constraint Language (SHACL)

W3C Recommendation

Abstract

This document defines the SHACL Shapes Constraint Language, a language for validating RDF graphs against a set of conditions. These conditions are provided as shapes and other constructs expressed in the form of an RDF graph. RDF graphs that are used in this manner are called "shapes graphs" in SHACL and the RDF graphs that are validated against a shapes graph are called "data graphs". As SHACL shape graphs are used to validate that data graphs satisfy a set of conditions they can also be viewed as a description of the data graphs that do satisfy these conditions. Such descriptions may be used for a variety of purposes beside validation, including user interface building, code generation and data integration.

Validation

Given a finished graph: **is it valid?**

The official RML shapes already answer this.

Authoring

Starting from nothing: **how do I build it?**

Which resource first, which fields, which choices.

Source: Shapes Constraint Language (SHACL) W3C Recommendation 20 June 2017. url:https://www.w3.org/TR/shacl/

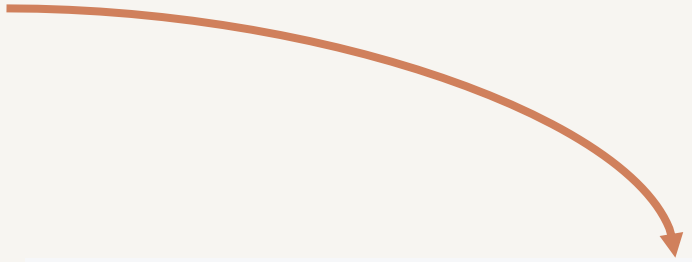
SHACL, an example

SHACL

Turtle serialization

```
@prefix sh:  <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix ex:  <http://example.org/> .

ex:PersonShape a sh:NodeShape ;
    sh:targetClass ex:Person ;     # applies to every ex:Person
    sh:property [
        sh:path ex:age ;
        sh:datatype xsd:integer ;    # the age is an integer
        sh:minCount 1 ;              # at least one age
        sh:maxCount 1 ;              # at most one age
        sh:minInclusive 0 ;          # not negative
    ] .
```



Generated form

Add node

http://example.org/PersonShape

IRI: http://example.org/personshape-1

* age

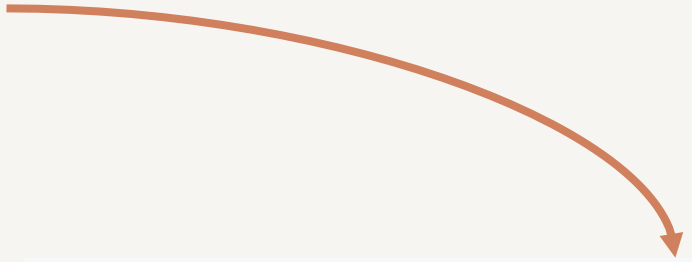
SHACL, an example

SHACL

Turtle serialization

```
@prefix sh:  <http://www.w3.org/ns/shacl#> .
@prefix xsd: <http://www.w3.org/2001/XMLSchema#> .
@prefix ex:  <http://example.org/> .

ex:PersonShape a sh:NodeShape ;
    sh:targetClass ex:Person ;     # applies to every ex:Person
    sh:property [
        sh:path ex:age ;
        sh:datatype xsd:integer ;    # the age is an integer
        sh:minCount 1 ;              # at least one age
        sh:maxCount 1 ;              # at most one age
        sh:minInclusive 0 ;          # not negative
    ] .
```



Generated form

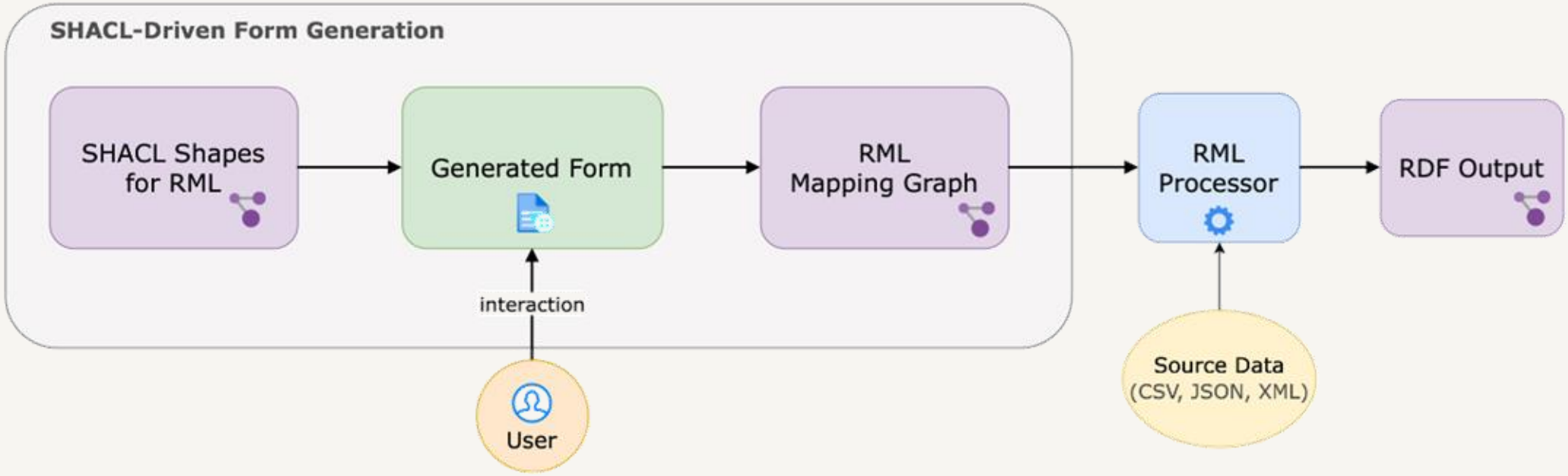
Add node

http://example.org/PersonShape

IRI: http://example.org/Carol

* age ⚠ -3 ✕

Workflow


SHACL-Driven Form Generation
SHACL Shapes for RML
Generated Form
RML Mapping Graph
RML Processor
RDF Output
interaction
User
Source Data
(CSV, JSON, XML)

Demo

A Prototype SHACL-Driven Form Generator for RML

(Demo)

https://christalisation.github.io/darmstadt-shacl-form/rml/

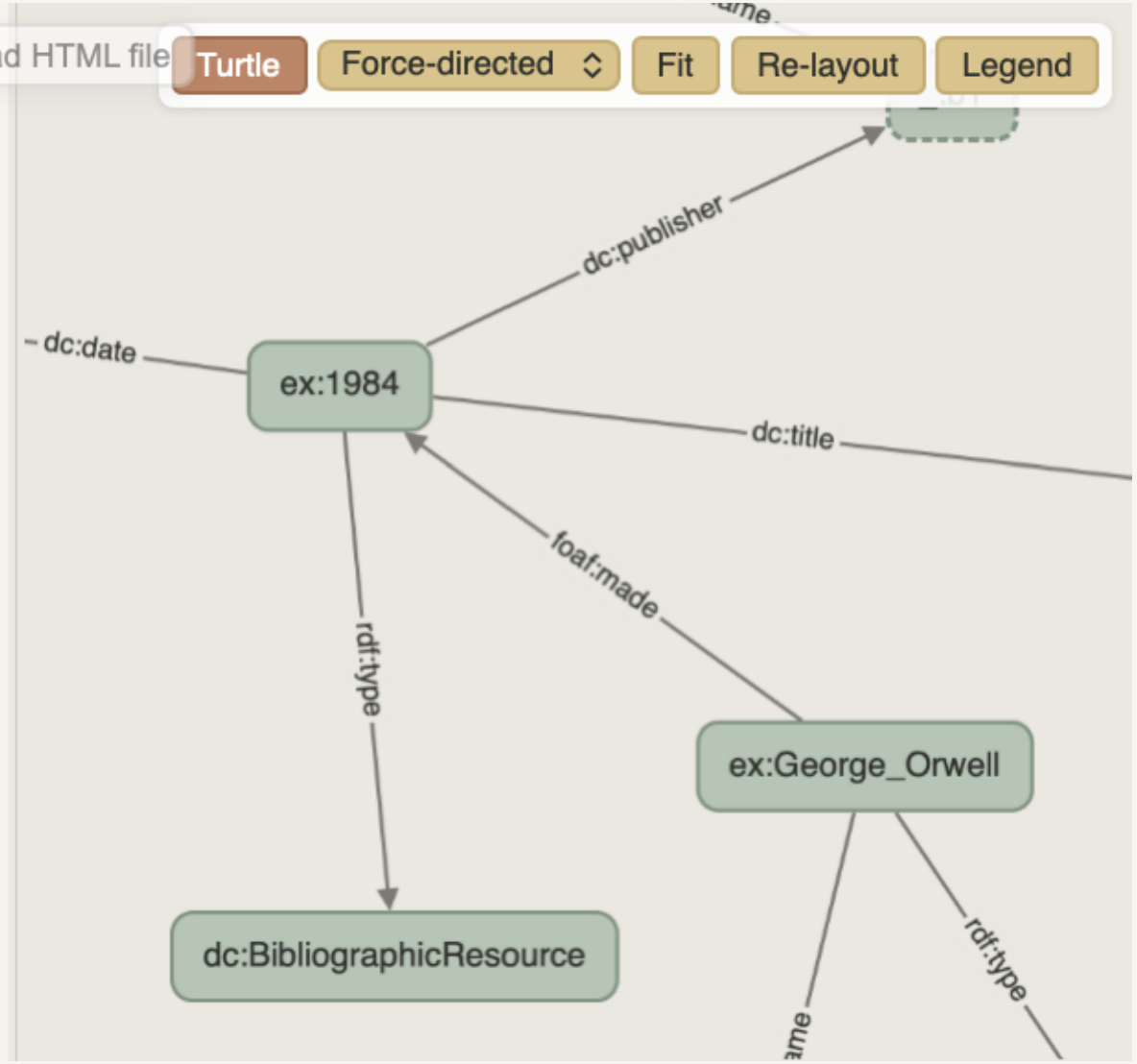
Demo: Output RDF viewer

```
@prefix ex:   <http://example.org/> .
@prefix foaf: <http://xmlns.com/foaf/0.1/> .
@prefix dc:   <http://purl.org/dc/elements/1.1/>
@prefix xsd:  <http://www.w3.org/2001/XMLSchema#

ex:George_Orwell a foaf:Person ;
  foaf:name "George Orwell" ;
  foaf:made ex:1984 .

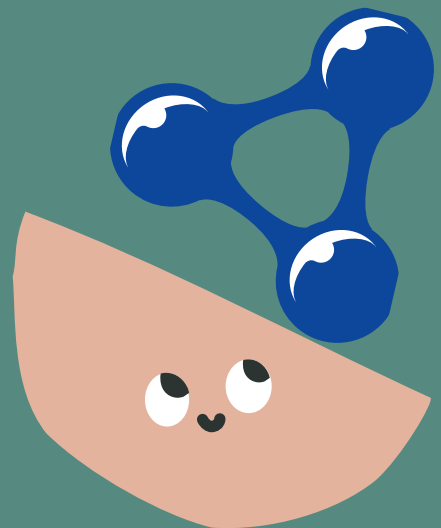
ex:1984 a dc:BibliographicResource ;
  dc:title "Nineteen Eighty-Four"@en ;
  dc:date "1949"^^xsd:gYear ;
  dc:publisher [ foaf:name "Secker & Warburg" ]
```

Render | Copy link | Ctrl+Enter



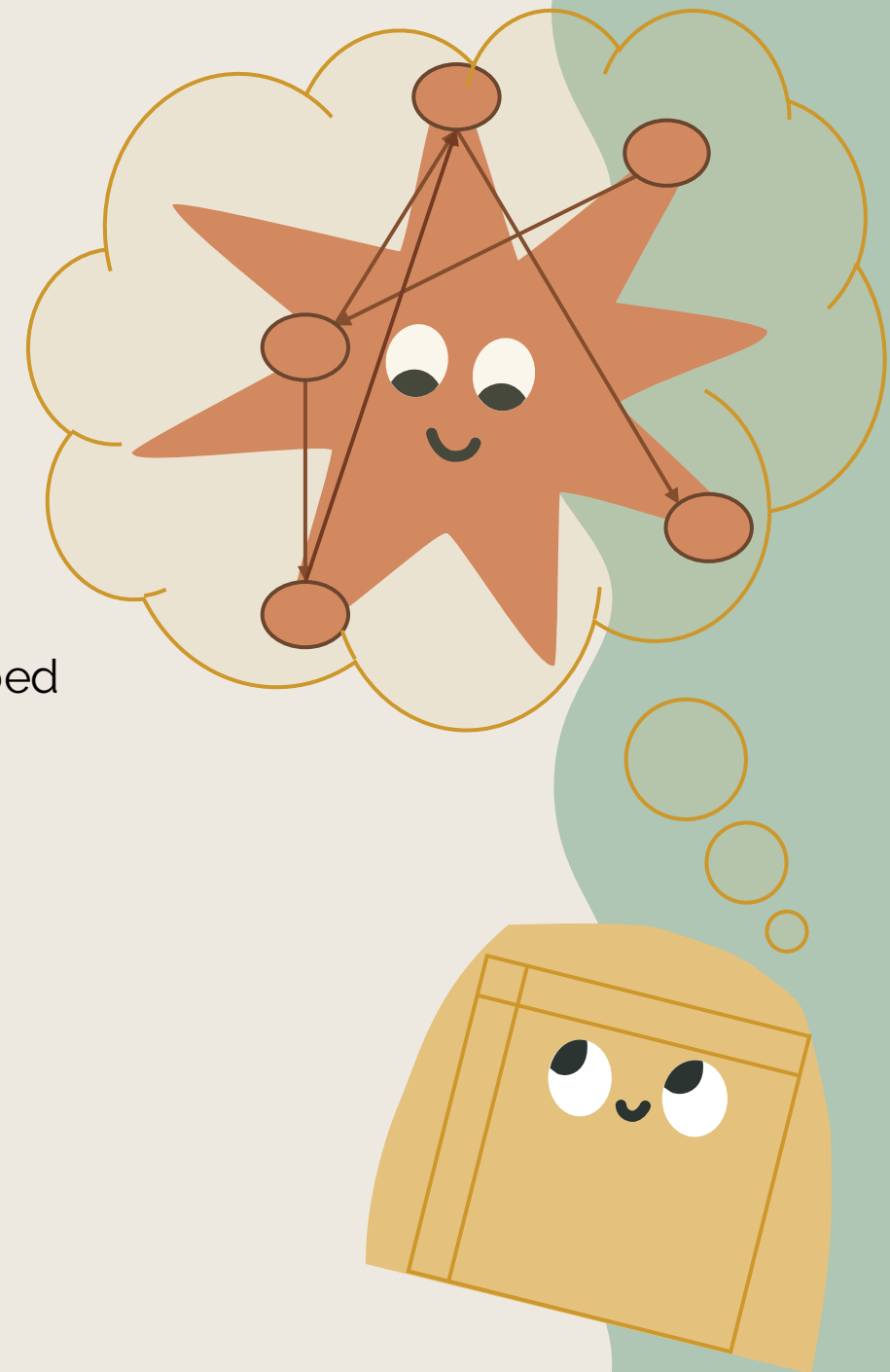
Some challenges...

1. Not every constraints become a field or a widget directly
2. SHACL says what is valid, not what can be build, nor how to build it.
   a) Needs decisions SHACL does not encode: entry points, node identity
   b) Needs extra requirements: reuse a created node, choose identifiers
3. How a shape is written changes the form you get.



Take-away

1. A specification says what is valid, can be described with SHACL
2. RML authouring through its SHACL shapes is possible, but needs added requirements.
3. When writing SHACL, it is worth thinking about authoring, not only about validation.



… Interested in testing the tool ?

christal.mangolopa@student.uliege.be

https://christalisation.github.io/darmstadt-shacl-form/

Thank you !

